Tzu-Chiao Lu (盧子喬)

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EDUCATION

2010 – 2017 Ph.D., Life Science, National Defense Medical Center, Taipei, Taiwan
2005 – 2008 M.S., Molecular Biology, National Chung Cheng University, Chiayi, Taiwan
2000 – 2004 B.S., Chemistry, National Chung Cheng University, Chiayi, Taiwan

RESEARCH EXPERIENCE

2026 – Assistant Professor, Institute of Biopharmaceutical Science, National Yang Ming Chiao Tung University
2024 – 2026 Instructor, Huffington Center on Aging & Department of Molecular and Human Genetics, Baylor College of Medicine, USA
2021 – 2024 Postdoctoral Fellow, Huffington Center on Aging & Department of Molecular and Human Genetics, Baylor College of Medicine, USA
2018 – 2021 Postdoctoral Fellow, National Yang Ming University, Taiwan
2010 – 2017 Doctoral Researcher, Academia Sinica, Taiwan
2009 – 2010 Research assistant, Academia Sinica, Taiwan
2005 – 2009 Master researcher and research assistant, National Chung Cheng University, Taiwan

AWARDS

2026 SEED Young Scholars Program
2026 Yushan Young Scholar Program
2023 Roy M. Huffington Distinguished Lecture & Estella E. Medrano Award for Excellence in Aging Research
2017 National Defense Medical Center Research Poster Outstanding Award
2017 SMBE Conference Travel Grant
2014 IBMS, Academia Sinica Conference Travel Grant

PUBLICATIONS

1. **TC Lu**, H Li. Clonal drift in the aging gut: Mapping the fate of stem cells over time. ***PLoS Biol.*** **2026**, 24(7):e3003928. PMID: 42490637
2. **TC Lu**[#], CY Liang[#], YJ Park, B Sun, T Jackson, J Chen, E Harrison, AL Hsu, Y Qi, H Li[*]. Rapamycin Mitigates a Sex-biased Convergent Aging Trajectory. ***bioRxiv*** **2026**.

3. NA Auld, YJ Park, T Jackson, S Eleraky, Y Younes, CY Liang, **TC Lu**, S Gong, Z Yin, B Sun, Y Zhang, TP Wu, Y Qi, H Li. Intrinsic regulation of intestinal stem cell fate and homeostasis by Tet. ***bioRxiv*** **2026**.
4. H Liu[#], Q Liu[#], JC Bean[#], Y Yang, Y Li, L Tu, M Yu, OZ Ginnard, KM Conde, M Wang, X Fang, H Liu, N Yin, Y Deng, S Jossy, HW Wong, M Burt, J Han, QT Tong, BR Arenkiel, M Xue, H Cui, **TC Lu**, Y Qi, H Li, C Wang, Y He, Y Xu[*]. Identification of a cold-sensing center in the hypothalamus. **2026 submitted.**
5. JC Bean, J Jian, **TC Lu**, H Liu, K Conde, DA Threat, SV Jossy, ME Burt, J Cheng, Y Deng, X Fang, X Geng, J Han, Y Li, H Liu, Q Liu, Y Liu, Y Shi, L Tu, M Wang, X Xu, Y Yang, M Yu, X Liu, M Sun, F Wang, OZ Ginnard, Y Yang, Y He, C Wang, Y Qi, H Li, Y Xu. Sex-specific differences in mediobasal hypothalamus in response to nutritional states. ***Nature Communications*** **2026**, 17(1):2941. PMID: 41714323.
6. GA Perez, Z Lai, GA Edwards III, JM Dundee, SN Leahy, C Qi, Y Qi, YJ Park, **TC Lu**, M Uddin, R Zhao, H Zheng, H Li, JL Jankowsky. Neuronal subtype governs amyloid structure, cellular response, and cognitive outcome in genetically targeted APP mouse models. ***Mol Neurodegener*** **2026**, 21(1):2. PMID: 41495836.
7. MP Marques, B Sun, YJ Park, T Jackson, **TC Lu**, Y Qi, E Harrison, MC Wang, K Venkatachalam, OM Pasha, A Varanasi, DK Carey, DR Mani, J Zirin, M Qadiri, Y Hu, N Perrimon, SA Carr, ND Udeshi, L Luo, J Li, H Li. In-situ glial cell-surface proteomics identifies pro-longevity factors in *Drosophila*. ***eLife*** **2026**. PMID: 42550595
8. YJ Park[#], **TC Lu[#]**, T Jackson, LD Goodman, L Ran, J Chen, CY Liang, E Harrison, C Ko, X Chen, B Wang, AL Hsu, E Ochoa, KF Bieniek, S Yamamoto, Y Zhu, H Zheng, Y Qi, HJ Bellen[*], H Li[*]. Distinct systemic impacts of Aβ42 and Tau revealed by whole-organism snRNA-seq. ***Neuron*** **2025**, 113:2065-2082. PMID: 40381615
9. M Qadiri, Y Liu, AR Kim, M Han, E Zhou, A Veal, **TC Lu**, H Li, Y Hu, N Perrimon. FlyPhoneDB2: A Computational Framework for Analyzing Cell-Cell Communication in *Drosophila* scRNA-seq Data Integrating AlphaFold-Multimer Predictions. ***Comput Struct Biotechnol J.*** **2025**, 27:2814-2822. PMID: 40677242
10. OV Goldman, AE DeFoe, Y Qi, Y Jiao, SC Weng, LH Houri-Zeevi, P Lakhiani, T Morita, J Razzauti, A Rosas-Villegas, YN Tsitohay, M Walker, BR Hopkins, Mosquito Cell Atlas Consortium, JXD Ang, I Antoshechkin, Y Cai, F Chen, YC Chen, J Devilliers, L Dong, R Feuda, P Gabrieli, A Kopp, H Kwon, HH Li, **TC Lu**, T Lucio, JT Marques, MF Oliveira, RP Olmo, U Palatini, ZM Pithawala, J Pompon, Y Reis, J Rodrigues, RC Smith, OS Akbari, LB Duvall, HW-Cooper, TR Sorrells, R Sharma, H Li[*], LB Vosshall[*], N Shai[*]. Mosquito Cell Atlas: A single-nucleus transcriptomic atlas of the adult Aedes aegypti mosquito. ***Cell*** **2025**, 188:7267-7290. PMID: 41172998
11. PC Hsu, **TC Lu**, PH Hung, JY Leu. Protein moonlighting by a target gene dominates phenotypic divergence of the Sef1 transcriptional regulatory network in yeasts. ***Nucleic Acids Research*** **2024**. 52:13914-13930. PMID: 39565215
12. B Wang, H Martini-Stoica, C Qi, **TC Lu**, S Wang, W Xiong, Y Qi, Y Xu, M Sardiello, H Li, H Zheng. TFEB-vacuolar ATPase signaling regulates lysosomal function and microglial activation in tauopathy. ***Nature Neuroscience*** **2024**. 27:48-62. PMID: 37985800
13. **TC Lu[#]**, M Brbić[#], YJ Park, T Jackson, J Chen, SS Kolluru, Y Qi, NS Katheder, XT Cai, S Lee, YC Chen, N Auld, D Welsch, S D'Souza, AO Pisco, RC Jones, J Leskovec, EC Lai, HJ Bellen, L Luo, H Jasper[*], SR Quake[*],

H Li*. Aging Fly Cell Atlas Identifies Exhaustive Aging Features at Cellular Resolution. ***Science*** **2023**, 380:eadg0934. PMID: 37319212

14. NS Katheder, KC Browder, D Chang, A De Maziere, P Kujala, S van Dijk, J Klumperman, **TC Lu**, H Li, Z Lai, D Sangaraju, H Jasper. Nicotinic acetylcholine receptor signaling maintains epithelial barrier integrity. ***Elife*** **2023**. 12:e86381. PMID: 38063293
15. A Mamiya, A Sustar, I Siwanowicz, Y Qi, **TC Lu**, P Gurung, C Chen, JS Phelps, AT Kuan, A Pacureanu, WC A Lee, H Li, N Mhatre, JC Tuthill. Biomechanical origins of proprioceptor feature selectivity and topographic maps in the *Drosophila* leg. ***Neuron*** **2023.** 111:3230-3243. PMID: 37562405
16. CY Lim, HT Lin, C Kumsta, **TC Lu**, FY Wang, YH Kang, M Hansen, TT Ching, AL Hsu. SAMS-1 coordinates HLH-30/TFEB and PHA-4/FOXA activities through histone methylation to mediate dietary restriction-induced autophagy and longevity. ***Autophagy*** **2023**, 19:224-240. PMID: 35503435
17. EM Crombie, S Kim, S Adamson, H Dong, **TC Lu**, Y Wu, Y Wu, Y Levy, N Stimple, WMR Lam, HW D Hey, DJ Withers, AL Hsu, BH Bay, J Ochala, SY Tsai. Activation of eIF4E-binding-protein-1 rescues mTORC1-induced sarcopenia by expanding lysosomal degradation capacity. ***Journal of Cachexia*** **2022**, Sarcopenia and Muscle. 14:198-213. PMID: 36398408
18. M Herre, OV Goldman, **TC Lu**, G Caballero-Vidal, Y Qi, ZN Gilbert, Z Gong, T Morita, S Rahiel, M Ghaninia, R Ignell, BJ Matthews, H Li, LB Vosshall, MA Younger. Non-canonical odor coding in the mosquito. ***Cell*** **2022**, 185:3104-3123. PMID: 35985288
19. H Li, J Janssens, M De Waegeneer, SS Kolluru, K Davie, V Gardeux, W Saelens, FPA David, M Brbić, …, **TC Lu**, …, N Perrimon, B Deplancke, SR Quake, L Luo, S Aerts. Fly Cell Atlas: A single-nucleus transcriptomic atlas of the adult fruit fly. ***Science*** **2022**, 375:eabk2432. PMID: 35239393
20. PC Hsu, **TC Lu**, PH Hung, YT Jhou, AAA Amine, CW Liao, JY Leu. Plastic rewiring of Sef1 transcriptional networks and the potential of non-functional transcription factor binding in facilitating adaptive evolution. ***Mol Biol Evol*** **2021**, 38:4732-4747. PMID: 34175931
21. S Sural, **TC Lu**, SA Jung, AL Hsu. HSB-1 Inhibition and HSF-1 Overexpression Trigger Overlapping Transcriptional Changes To Promote Longevity in *Caenorhabditis elegans*. ***G3 (Bethesda)*** **2019**, 9:1679-1692. PMID: 30894454
22. **TC Lu**, JY Leu, WC Lin. A Comprehensive Analysis of Transcript-Supported *De Novo* Genes in Saccharomyces sensu stricto Yeasts. ***Mol Biol Evol*** **2017**, 34:2823-2838. PMID: 28981695

MEETING ORGANIZATION

2025 66th Annual *Drosophila* Research Conference, Techniques and Technology Session (co-chair)

2024 65th Annual *Drosophila* Research Conference, Techniques and Technology Session (co-chair)

INVITED TALKS

2026 April 26 Department of Developmental Biology, Washington University in St. Louis

2026 March 27 Center for Molecular Psychiatry, University of South Florida

2025 August 1 Institute of Biopharmaceutical Sciences, National Yang Ming Chiao Tung University, Taipei, Taiwan (virtual)

2025 March 21 66th Annual *Drosophila* Research Conference, San Diego, USA

2025 January 21 Institute of Biological Chemistry, Academia Sinica, Taipei, Taiwan
2024 May 15-16 BIOTOOLS seminar, Taiwan (virtual)
2023 August 10 PharmiGENE seminar, Taiwan (virtual)
2023 June 21 College of Life Sciences, National Yang Ming Chiao Tung University, Taipei, Taiwan
2023 June 15 Institute of Biomedical Sciences, Academia Sinica, Taipei, Taiwan
2023 June 9 Department of Biomedical Sciences, National Chung Cheng University, Chiayi, Taiwan
2023 June 7 Department of Biological Science and Technology, National Yang Ming Chiao Tung University, Hsinchu, Taiwan
2023 March 2 64th Annual *Drosophila* Research Conference, Chicago, USA
2020 December 14 HGT Bioinformatics Symposium and Workshop, Taipei, Taiwan

ACADEMIC SERVICE

Reviewer for Journals: STAR Protocols, Cell & Bioscience

MENTORING AND TEACHING

Teaching assistant:

- "Introduction to Life Science", Department of Life Science, National Chung Cheng University, 2006 Fall (class size ~20)

Computational Biology Mentor: Designed independent trainee projects and provided domain background; guided trainees to (1) translate biological questions into testable hypotheses and analysis plans, (2) run reproducible pipelines on HPC/SLURM with containerized environments (Conda/Bioconda), (3) perform QC (e.g., doublet/ambient RNA filtering), integration/batch correction (Harmony), clustering/trajectory inference (Leiden/PAGA), differential and pathway analysis, and publication-grade visualization (tSNE/UMAP, dot/violin plots); (4) present results (talks/posters) and document methods; project management, and career development.

Hongjie Li's Lab, Baylor College of Medicine

- Niccole Auld, graduate student, Jul 2021-2026
- Ye-Jin Park, graduate student, Sep 2021-2026
- Tyler Jackson, graduate student, Jan 2022-2026
- Erin Harrison, undergraduate student from Rice University, and research assistant, Sep 2022-Jun 2025
- Doug Welsch, undergraduate student from Rice University, Sep 2022-Jan 2024
- Jiaye Chen, graduate student on rotation, Aug 2022-Oct 2022
- Yang Zhang, graduate student on rotation, Oct 2022-Dec 2022
- Ryan Cheng, graduate student on rotation, Aug 2023-Oct 2023
- Tzu-Hsuan Huang, graduate student on rotation, Oct 2023-Dec 2023
- Justin Dhindsa, graduate student on rotation, Aug 2024-Oct 2024

Jeannie Chin's lab, Baylor College of Medicine

- Yoshitaka Furuta, graduate student, Apr 2024-2026

Vishnu Anand Cuddapah's lab, Baylor College of Medicine

- Bhargavi Gundamalla, bioinformatics analyst, Apr 2025-2026

Ao-lin Hsu's Lab, University of Michigan, National Yang-Ming University

- Surojit Sural, graduate student, Jan 2018-Oct 2020

Wen-chang Lin's lab, Academia Sinica, Taiwan

- Wei-Ting Kuo, master student from National Yang-Ming University, Sep 2015-Jul 2016

Jun-Yi Leu's lab, Academia Sinica, Taiwan

- Po-Hsiang Hung, graduate student, Sep 2011-Sep 2017
- Chia-Wei Liao, research assistant, Sep 2015-Sep 2017

Bench Mentor: Designed independent trainee projects; taught relevant scientific background and assay execution; guided trainees in (1) formulating questions and designing controlled experiments, (2) executing validated protocols, (3) organizing, compiling, and analyzing data, and (4) presenting results to scientific audiences; additionally advised on career options and professional development.

Jun-Yi Leu's lab, Academia Sinica, Taiwan

- Po-Hsiang Hung, graduate student, Sep 2011-Sep 2017
- Chia-Wei Liao, research assistant, Sep 2015-Sep 2017

Hau-Ren Chen's Lab, National Chung Cheng University, Taiwan

- Sim-kun Ng, master student, Sep 2006-Jun 2009
- Wei-Shen Wu, master student, Sep 2006-Jun 2009
- Yu-Ting Kao, undergraduate student, Sep 2005-Jun 2007
- Yu-wen Huang, undergraduate and master student, Sep 2006-Jun 2008

***Lab workshops*:**

- "Data to Figure: Adobe illustrator workshop", Li Lab, May 2025.
- "Hands-on single-cell data using CellxGene", Li Lab, Karpac Lab, September 2023.
- "Introduction to single-cell RNA sequencing", Hsu Lab, February 2019.
- "Everything you want to know about VectorNTI", Lo Lab, September 2009.

Local bioinformatics workshop:

- "Introduction to single-cell RNA sequencing", Taipei Bioinformatics Omnibus, March 8th, 2019 (workshop size ~20)

THESIS COMMITTEE MEMBER

Yi-Tzu Lo, M.S., National Yang Ming Chiao Tung University, 2024